

Kumar Kaustuk Raj

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github.com/Manasdbg123 | [Portfolio](#) | [Codeforces](#) | [LeetCode](#)

SUMMARY

Software Developer building AI-driven backend systems, including agentic LLM workflows for enterprise automation, an AI-powered Learning Management System (LMS) using FastAPI and PostgreSQL, and a computer-vision-based Automatic Number Plate Recognition (ANPR) pipeline. Skilled in Java, Spring Boot, Python, FastAPI, REST API design, Spring Security, JWT, PostgreSQL, MySQL, Microservices, LLM Agents, and system architecture.

EDUCATION

National Institute of Technology, Andhra Pradesh Dec 2021 – Apr 2025
Bachelor of Technology in Computer Science and Engineering Andhra Pradesh, India

EXPERIENCE

Wittybrains Software Technologies Pvt. Ltd. Jul 2026 – Present
Software Developer Noida, India

- Architected an **Agentic ERP** module using role-aligned LLM agents (planner–executor–reflector pattern) integrated with existing ERP APIs and a risk-tiered human-in-the-loop review step, automating inventory and order-processing workflows across **3+** business modules and cutting manual intervention by **35%**; also built an **AI-powered LMS** from scratch using **FastAPI** and **PostgreSQL** with adaptive learning paths, AI-generated quizzes, and a RAG-based tutor chatbot, backed by **20+** RESTful APIs serving **500+** learners across **40+** courses.
- Developed an **ANPR** pipeline using YOLO-based plate detection and OCR (EasyOCR/PaddleOCR) to identify vehicle registration numbers, with a classification module for vehicle make, model, and color from camera feeds; applied image preprocessing (adaptive thresholding, denoising, contrast enhancement) to improve OCR accuracy under varying lighting, and deployed the pipeline via FastAPI for real-time inference on live and recorded video streams.

ACHIEVEMENTS

- Codeforces:** Rated **Candidate Master (1913)**; **Global Rank 57** in Codeforces Round 1030 (Div. 2).
- LeetCode:** Rated **Guardian (2183)**; **Global Rank 11** in a Weekly Contest.
- Competitive Programming:** Solved **2400+** problems across Codeforces, LeetCode, GeeksforGeeks, InterviewBit.

PROJECTS

Food Delivery Platform GitHub Jan 2025 – May 2025

- Architected an event-driven microservices backend using Java and Spring Boot with **Apache Kafka** for async communication between catalog, user, and order services.
- Integrated Redis for data caching and session management, reducing query latency by **40%** and improving throughput.
- Developed **15+** secure RESTful APIs for client-service communication, authenticated via Spring Security and JWT.
- Engineered transactional integrity for high-volume order placements using Spring Data JPA and MySQL.
- Built an **LLM agent** for automated order-status and complaint handling, and a **RAG-based** recommendation engine for personalized order suggestions.
- Tech:** Java, Spring Boot, Microservices, Apache Kafka, Redis, REST APIs, Spring Security, JWT, Hibernate, MySQL, LLM Agents, RAG

NestXchange – Multi-Category Marketplace GitHub Feb 2023 – Present

- Built a full-stack marketplace for property and vehicle rent/buy/sell listings, with secure browsing, filtering, and landlord/seller communication.
- Designed a category-agnostic Listing entity with a JSONB-based schema in PostgreSQL, sharing one data model across categories.
- Built a unified search engine filtering universal fields (price, location, mode) and category-specific attributes via dynamic JSONB queries and a schema registry.
- Implemented a shared transaction/state-machine lifecycle (Available, Requested, Confirmed, Active/Completed, Closed) with pluggable rent/sale handlers.
- Built an **LLM agent** for conversational listing search and a **RAG-based** natural-language query engine over listing data.
- Configured access control and identity verification via Spring Security and JWT.
- Tech:** Java, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, PostgreSQL (JSONB), React, MySQL, JavaScript, HTML, CSS, LLM Agents, RAG

SKILLS

Languages: Java, Python, HTML, CSS, JavaScript, C++, SQL
Frameworks & APIs: Spring Boot, FastAPI, Microservices, RESTful APIs, Spring Security, JWT, Spring Data JPA, Hibernate, Servlets, JSP, JDBC, JavaMail API
AI/ML & Computer Vision: LLM Agents, RAG (Retrieval-Augmented Generation), OpenCV, YOLO, OCR, EasyOCR, PaddleOCR, Gemini AI
Databases & Caching: PostgreSQL, JSONB, MySQL, MongoDB, Redis
Tools & Cloud: Apache Kafka, Apache Tika, AWS, Git, Maven, Tomcat, Docker, Postman